

Intelligent Agent :-

1. Ability to interact with the real world.

→ to perceive

like

sensors can sense

→ pressure sensor can sense pressure etc.

→ speech recognition & understanding

→ image understanding

→ ability to take action.

2. Knowledge representation :-

→ Reasoning & Learning

→ modelling the external world

→ solving the problem

3. Learning & Adaptation :-

→ continuously learning & adapting

→ internal model are always being updated.

Subject: Vacuum Cleaner Agent:-

Percept sequence

Action

[A, clean]

Right

[A, dirty]

Suck

[B, clean]

Left

[B, dirty]

Suck

Environment types:-

- (i) Observable or non-observable
- (ii) Deterministic or non-deterministic
- (iii) Static or Dynamic.
- (iv) Discrete or Continuous.

Types of Agent:-

(i) Simple reflex agent

→ if car in my front
braking, agent will break

(iii) Reflex agents with state → They will track the states.

Subject: (iii) Goal based agent $\rightarrow$ Work until goal is achieved

(iv) Utility based agents $\rightarrow$ Map a state to a real function.